YouTube Trending Video Analytics

Abstract:

This project analyzes YouTube trending video data across different regions to identify patterns in content performance. The dataset includes features such as category, views, likes, comments, country, and publish time. Data preprocessing and sentiment analysis were performed using Python, and SQL was used to rank categories based on average views. The results were visualized using a dashboard to uncover insights related to popular genres, audience engagement, and regional differences. The project helps understand what type of content performs best on YouTube.

Introduction:

YouTube is one of the largest video-sharing platforms, where trending videos reflect current user interests and content popularity. Understanding the factors that make a video trend can help content creators and businesses improve their strategies.

This project focuses on analyzing trending video datasets from multiple regions to discover patterns in views, engagement, and sentiment. By using data analytics techniques, we aim to identify which categories are most popular, how audience engagement varies, and how trends differ across countries.

Methodology:

The dataset was first cleaned by removing duplicates and handling missing values. Date columns were converted into proper formats, and a new feature called *days_to_trend* was created to measure how quickly a video becomes trending.

Sentiment analysis was performed on video titles to classify them as positive, negative, or neutral. SQL queries were used to rank categories based on average views. Finally, the processed data was visualized using charts and dashboards to identify meaningful patterns.

Key Patterns / Insights:

- **Popular Categories:** Entertainment and Music videos receive the highest number of views, making them the most trending categories.
- **Regional Differences:** Trending patterns vary across countries, with some regions showing higher engagement levels than others.
- **Sentiment Impact:** Videos with positive sentiment titles tend to attract more views and engagement.
- **Engagement Relationship:** Videos with higher likes and comments generally have higher view counts.
- **Trending Speed:** Videos that trend quickly after publishing tend to perform better overall.

Tools and Technologies Used:

- Python (Pandas, Matplotlib, Seaborn) – Data cleaning and analysis
- TextBlob – Sentiment analysis
- SQL – Data querying and category ranking
- Tableau – Dashboard and visualization

Conclusion:

This project successfully identifies patterns in YouTube trending videos using data analytics techniques. The analysis shows that content category, sentiment, and engagement significantly influence video performance. These insights can help content creators and marketers design better strategies to increase reach and audience engagement. Overall, the project demonstrates how data-driven approaches can be used to understand trends in digital platforms.